SIX WEEKS INDUSTRIAL TRAINING

PROJECT REPORT

ON

"LOAN APPROVAL PREDICTION SYSTEM"

Submitted in partial fulfillment of the requirement

for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE & ENGINEERING



Submitted By: -

Chandan Ram
2447931
Data Science & Machine Learning
Session: (June-July)2026

Submitted To: -

Dr. Rajesh
H.O.D CSE DEPT

CERTIFICATE BY COMPANY

DECLAERATION

I hereby declare that the Industrial Training Report entitled "LOAN APPROVAL PREDICTION SYSTEM" is an authentic record of my own work carried out at **Solitaire Infosys Pvt. Ltd.** as requirement of 6-weeks Industrial Training during the period from June 2026 to July2026 for the award of degree of **B.Tech (CSE), Quest Group Of Institutions MOHALI** under the guidance of **Mr. Arshdeep Singh .**

CHANDAN RAM

(2447931)

Date: _____

Certified that the above statement made by the student is correct to the best of our knowledge and belief.

_____

Industrial Supervisor
(Signature and Seal)

ACKNOWLEDGEMENT

Through this acknowledgement, I express my sincere gratitude to all those people who have been associated with this project and have helped me with it and made it a worthwhile experience. First and foremost, I would like to thank **Solitaire Infosys** for giving me courage to make this project. At the outset, I would like to propose a word of thanks for the people who gave me unending support and help in numerous ways.

Firstly, I express thanks to **Dr. Rajesh (**H.O.D CSE DEPT) who gave me this opportunity to learn the subject in a particular approach who guided me and gave me valuable suggestions regarding the project report.

Secondly, I would like to thanks **Mr. Arshdeep Singh** and FACULTY MEMBERS of **Solitaire Infosys** who gave me their immense support in completing the project. The atmosphere provided was full of gaining more and more knowledge and to keep enthusiastic nature. The teachers provided a lot of help in resolving our doubts and making this project successful in shorter time.

Lastly, I would also like to thank my parents and friends who have helped me a lot in finishing the project in limited time.

I make this project not only for marks but also to increase my knowledge. Thanks again to all who helped me in completion of this project.

CHANDAN RAM
5TH Sem
B.Tech. (CSE)
Quest Group of Institutions

ABSTRACT

Loan approval is a critical decision for financial institutions. Approving loans without proper evaluation can lead to defaults, while rejecting good applicants causes revenue loss. Manual evaluation is time-consuming and prone to inconsistency. Automating this process using machine learning can lead to faster, fairer, and more consistent decisions.

The objective of this project is to develop a machine learning-based Loan Approval Prediction System. The system predicts whether a loan application is likely to be approved or rejected by analyzing applicant-related factors such as gender, marital status, number of dependents, education, employment status, income, co-applicant income, loan amount, loan term, credit history, and property area.

The project uses a dataset containing 614 historical loan application records. A Random Forest Classifier was trained using an 80:20 train-test split. The model achieved 86.2% accuracy, 84.0% precision, 98.8% recall, and 90.8% F1 score on test data.

The final model was deployed using Streamlit. The application allows users to enter applicant details and receive a loan approval prediction along with a confidence score and feature importance analysis.

Keywords

Machine Learning, Loan Approval Prediction, Credit Risk, Random Forest, Streamlit, Financial Analytics

ABOUT THE COMPANY

Solitaire Infosys Pvt. Ltd. is an acclaimed IT service provider contributing its part in the development of many businesses around the globe. We socialize with our clients to get a superior cognizance of their business and requirements and help them in fabricating websites and applications for their business. Founded in 2011 by a dynamic duo with the same aim and zeal, we have come a long way in satisfying our clients.

We are serving our clients with the world-class services for more than seven years now. The clients are delivered with the best IT solutions after we have developed a great understanding of their business and requirements. Our team works on the client projects like its own and that is the reason why we hold the edge in the league.

With every project that we deliver, we deliver our respect, creativity, quality, transparency, and teamwork to our clients. We have the experience, expertise, and capabilities to enable organizations to accelerate their service processes in every possible way. We are known for our excellent customer satisfaction, cost-effectiveness, and innovative skills that are unparalleled.

Services: -

1. Mobile Applications
2. Web Development
3. Digital Marketing
4. UX & UI Design
5. Enterprise Applications

TABLE OF CONTENTS

Chapter No.	Topic	Page No.
1	INTRODUCTION	8-9
2	PROJECT PROPOSAL	10
3	PROBLEM STATEMENT AND BJECTIVES	11
4	DATASET DESCRIPTION	12
5	DATA CLEANING REPORT	13
6	EXPLORATORY DATA ANALYSIS	14
7	FEATURE ENGINEERING REPORT	15
8	METHODOLOGY	16
9	MODEL TRAINING REPORT	17
10	MODEL EVALUATION REPORT	18
11	RESULT ANALYSIS	19
12	DEPLOYMENT REPORT	20
13	TESTING REPORT	21
14	CONCLUSION AND FUTURE SCOPE	22
15	REFERENCES AND APPENDIX	23-26

CHAPTER- 1
INTRODUCTION

1. Introduction

Artificial Intelligence (AI) and Machine Learning (ML) have transformed the way organizations make decisions by analysing data and identifying meaningful patterns. These technologies are widely used in industries such as healthcare, banking, finance, education, agriculture, and retail to improve accuracy, efficiency, and productivity. In the banking sector, machine learning has become an important tool for automating the loan approval process, enabling financial institutions to make faster and more consistent lending decisions.

Traditionally, loan applications are evaluated manually based on factors such as income, employment status, credit history, loan amount, and repayment period. This process is often time-consuming and may be affected by human judgement or inconsistencies. Machine learning addresses these challenges by learning from historical loan data and predicting whether a new application is likely to be approved or rejected.

The Loan Approval Prediction System developed in this project uses a Random Forest Classifier to predict loan approval based on applicant details, including gender, marital status, education, employment status, applicant and co-applicant income, loan amount, loan term, credit history, and property area. The trained model is deployed using Streamlit, providing a simple and interactive web interface where users can enter applicant information and receive instant loan approval predictions. This system improves the efficiency, accuracy, and reliability of the loan evaluation process.

Background

Traditional loan approval relies on loan officers reviewing applications and making decisions based on policies, financial details, and personal judgement. While this works well for simple cases, it becomes difficult when applications include multiple moderate risk factors. Different reviewers may assess the same information differently, leading to inconsistent decisions. Machine learning solves this problem by learning from historical loan data and identifying how each feature affects loan outcomes. As a result, it can provide fair, consistent, and data-driven loan approval predictions, improving both accuracy and efficiency.

Importance of the Project:

- Helps banks identify risky loan applicants before approving loans.
- Makes loan approval decisions more accurate and consistent.
- Reduces the time and effort required for manual loan verification.
- Supports loan officers by providing a data-based prediction.
- Minimizes human errors and personal bias in decision-making.
- Speeds up the overall loan approval process.

- Improves customer satisfaction by providing faster results.
- Demonstrates the practical use of Machine Learning in the banking sector.

Real-Life Use:

The Loan Approval Prediction System can be used by:

- Commercial banks
- Non-Banking Financial Companies (NBFCs)
- Credit unions
- Microfinance institutions
- Digital lending platforms
- Financial technology (FinTech) companies
- Credit risk assessment teams

The system serves as a decision-support tool by providing an initial prediction that helps institutions process loan applications more efficiently.

Scope of the report:

This report presents the complete development of the Loan Approval Prediction System carried out during the six-week Industrial Training. It covers dataset understanding, data cleaning, exploratory data analysis, feature engineering, data preprocessing using a Scikit-learn pipeline, training and evaluation of a Random Forest Classifier, and performance analysis using accuracy, precision, recall, and F1-score. The trained model is deployed as an interactive Streamlit web application that enables users to enter applicant details and obtain instant loan approval predictions. Finally, the report discusses the system's limitations, ethical considerations, and potential future enhancements.

\

CHAPTER-2

PROJECT PROPOSAL

Project Proposal Report

Field	Details
Project Title	Loan Approval Prediction AI
Domain	Finance / AI / ML / Credit Risk Analytics
Problem Type	Binary Classification
Deployment	Streamlit Web Application

Problem Statement

Financial institutions often face difficulty in evaluating loan applications consistently. Loan defaults occur when high-risk applicants are approved, and revenue is lost when creditworthy applicants are rejected. Manual evaluation is slow and inconsistent.

Objectives

1. Load and prepare the historical loan application dataset (614 records).
2. Clean and handle missing values in the dataset.
3. Analyze important loan approval factors through EDA.
4. Train a Random Forest classification model for loan approval prediction.
5. Evaluate the model using accuracy, precision, recall, and F1 score.
6. Deploy the model using a Streamlit web application.
7. Provide loan approval prediction with a confidence score and feature analysis.

Technologies Used

Category	Technology
Language	Python
Libraries	Pandas, NumPy, Scikit-learn, Joblib
Visualization	Plotly, Matplotlib
Deployment	Streamlit
Data Storage	CSV

CHAPTER-3

PROBLEM STATEMENT AND OBJECTIVES

Problem Statement and Objectives

Loan approval decisions are influenced by financial, demographic, and credit-related factors. These include credit history, applicant income, co-applicant income, loan amount, loan repayment term, employment status, education level, marital status, number of dependents, and property area type.

The main problem addressed in this project is: How can machine learning be used to predict loan approval outcomes using available applicant financial indicators?

Need of the Project

Manual evaluation of every loan application is time-consuming and subject to human bias. Machine learning can process multiple applicants features simultaneously and generate a reliable approval prediction quickly.

Main Objectives

Objective	Description
Data Understanding	Understand the structure and features of the dataset.
Data Cleaning	Validate data ranges and prepare clean data.
Data Visualization	Identify trends using charts and graphs.
Model Training	Train a classification model.
Model Evaluation	Evaluate using accuracy, precision, recall, F1, and confusion matrix.
Deployment	Create a user-friendly Streamlit app.

Expected Outcome

The expected outcome is a web application that predicts whether a loan application will be Approved or Rejected, along with a confidence score.

Assumptions and Constraints

Assumptions

- The dataset is assumed to represent real loan applicants and is suitable for training the model.
- The model predicts loan approval only at the time of application.
- It is assumed that the applicant information entered is correct and complete.

Constraints

- The model is trained on historical data and does not use real-time loan information.
- The prediction accuracy depends on the quality and size of the dataset.
- Model training and testing were performed on a standard personal computer, limiting the use of large datasets and extensive hyperparameter tuning.

CHAPTER-4

DATASET DESCRIPTION

Dataset Description Report

The dataset is sourced from historical loan application records. It represents real-world loan applicant data commonly used in credit risk and financial analytics.

Description	Value
Number of Records	614
Total Columns	13
Model Input Features	6
Target Variable	Loan_Status
Problem Type	Binary Classification

Key Feature Description

Feature	Data Type	Description
Loan_ID	Categorical	Unique identifier for each loan application
Gender	Categorical	Applicant's gender (Male/Female)
Married	Categorical	Marital status of the applicant
Dependents	Categorical	Number of dependents of the applicant
Education	Categorical	Education level (Graduate/Not Graduate)
Self_Employed	Categorical	Whether the applicant is self-employed
ApplicantIncome	Float	Monthly income of the applicant
CoapplicantIncome	Float	Monthly income of the co-applicant
LoanAmount	Float	Loan amount requested by the applicant
Loan_Amount_Term	Integer	Loan repayment term in months
Credit_History	Binary	Credit history status (1 = Good, 0 = Poor)
Property_Area	Categorical	Location of the property (Urban, Semiurban, Rural)
LoanAmount_log	Float	Log-transformed loan amount used for model improvement
TotalIncome	Float	Combined income of applicant and co-applicant
TotalIncome_log	Float	Log-transformed total income used for model improvement
Loan_Status	Binary	Loan approval status (1 = Approved, 0 = Rejected)

Target Variable

Class	Meaning
0	Approved (Y)
1	Rejected (N)

CHAPTER-5

DATA CLEANING REPORT

Data Cleaning Report

Check	Value
Initial Rows	614
Initial Columns	13
Missing Values	0
Duplicate Records	0

Data Cleaning Steps Performed

1. Loaded raw loan application data from loan.csv.
2. Checked for duplicate records in the dataset.
3. Verified numerical columns and their valid ranges.
4. Filled missing categorical values (Gender, Married, Dependents, Self_Employed) using mode imputation.
5. Filled missing numerical values (LoanAmount with mean, Loan_Amount_Term and Credit_History with mode).
6. Removed Loan_ID as it is an identifier column not useful for prediction.
7. Engineered a new feature TotalIncome_log from ApplicantIncome and CoapplicantIncome.

Cleaning Example

Column	Check	Action Taken
Gender	Range validation	Kept within valid percentage range
Married	Range validation	Y/N
Loan_ID	Identifier column	Removed from model training
Loan_Status	Target variable	Encoded as 0 and 1

Final Dataset Shape

Rows: 614 | Columns: 13 (plus engineered TotalIncome_log)

CHAPTER-6

EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis Report

The objective of EDA is to understand loan approval patterns and study the relationship between applicant features and the loan approval outcome.

Visualizations Used

- Loan Status Distribution Chart
- Income and Loan Amount Distribution Analysis
- Credit History vs. Loan Status Analysis
- Property Area and Education Breakdown Chart

Key Observations

1. Good credit history (Credit_History = 1) is strongly associated with loan approval.
2. Higher applicant income and co-applicant income increase chances of loan approval.
3. Loan amount distributions are right-skewed; log transformation was applied to TotalIncome.
4. Semiurban property area applicants have a higher loan approval rate compared to Rural and Urban.
5. Graduates have a higher approval rate than non-graduates.

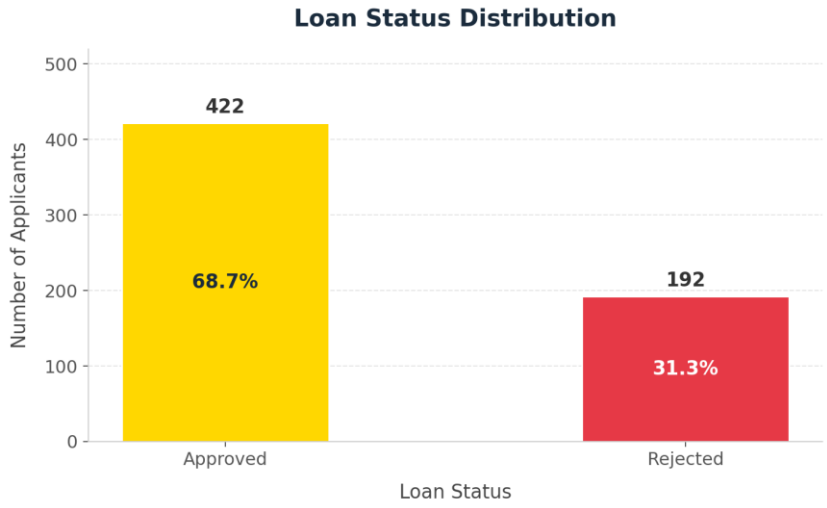


Figure 1: Loan Status Distribution (Approved vs Rejected)

CHAPTER-7

FEATURE ENGINEERING REPORT

7. Feature Engineering Report

Feature engineering is the process of transforming raw data into useful input features for machine learning models. It helps the model learn better patterns from structured data.

Techniques Applied

1. Numerical feature scaling using StandardScaler.
2. Categorical feature encoding using OneHotEncoder.
3. Target variable separation before model training.
4. ColumnTransformer pipeline creation for clean preprocessing.

Feature Groups

Group	Features
Numerical Features	Gender, Married, Dependents, Education, Self_Employed, Property_Area, ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term, Credit_History, TotalIncome_log
Target Feature	Loan_Status

Feature Engineering Summary

Numerical features were scaled using StandardScaler, categorical features were encoded using OneHotEncoder, and a log-transformed TotalIncome feature was engineered. All transformations were combined using a Scikit-learn ColumnTransformer pipeline.

CHAPTER-8

METHODOLOGY

Project Methodology

The project followed a complete machine learning development lifecycle. Each step was completed carefully to ensure that the final model was usable, understandable, and deployable.

Project Workflow

1. Problem Identification
2. Dataset Loading and Inspection
3. Data Validation and Cleaning
4. Exploratory Data Analysis
5. Feature Selection and Preprocessing
6. Train-Test Split
7. Model Training
8. Model Evaluation
9. Model Saving
10. Streamlit Deployment
11. Testing and Documentation

Machine Learning Approach

This project uses supervised machine learning because the dataset contains input features and a known target variable named Loan_Status. Since the output has two classes (Approved/Rejected), this is a binary classification problem.

Algorithm Used

The final algorithm used is Random Forest Classifier (n_estimators=200, max_depth=6). It was selected because it handles both numerical and categorical data well, reduces overfitting through ensemble learning, and provides interpretable feature importance values.

CHAPTER-9

MODEL TRAINING

Model Training Report

Dataset Part	Value
Training Data	491
Testing Data	123
Split Ratio	80:20

Model Pipeline

Component	Description
Numerical Transformer	StandardScaler
Categorical Transformer	OneHotEncoder
Model	RandomForestClassifier
Model Saving	Joblib

Random Forest Classifier

Random Forest is an ensemble learning algorithm that builds multiple decision trees and combines their predictions. It is useful for classification problems and performs well on structured tabular datasets.

Hyperparameters Used

Parameter	Value
n_estimators	200
max_depth	6
min_samples_split	2 (default)
min_samples_leaf	1 (default)
random_state	42
class_weight	balanced

The trained model was saved as models/loan_approval_model.joblib for use inside the Streamlit application.

CHAPTER-10

MODEL EVALUATION REPORT

10. Model Evaluation Report

Metric	Description
Accuracy	Measures total correct predictions
Precision	Correctness of predicted at-risk cases
Recall	How many actual at-risk cases were detected
F1 Score	Balance between precision and recall

Final Evaluation Metrics

Metric	Value
Accuracy	86.2%
Precision	84.0%
Recall	98.8%
F1 Score	90.8%

Confusion Matrix

Actual / Predicted	Predicted Rejected (N)	Predicted Approved (Y)
Actual Rejected (N)	14	24
Actual Approved (Y)	2	83

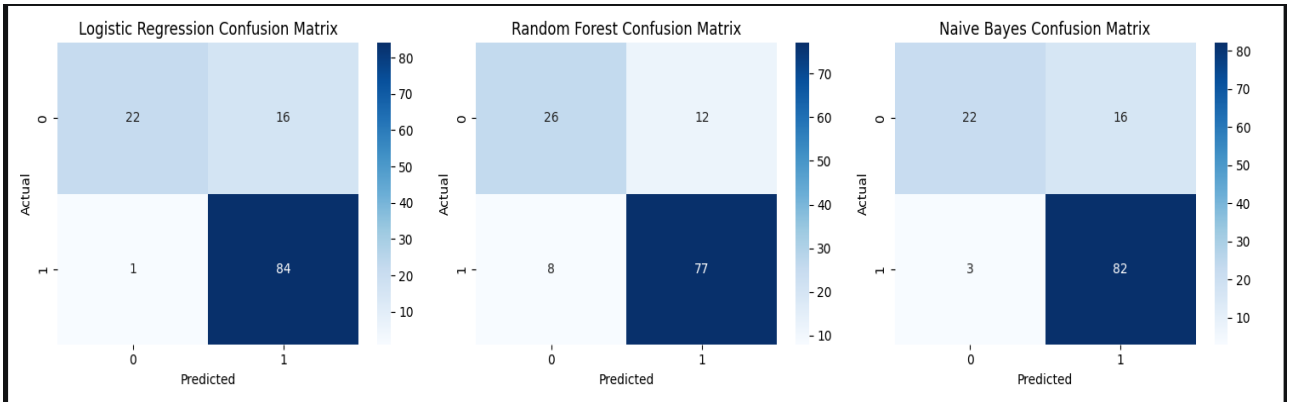


Figure 2: Confusion Matrix

The model achieved strong performance on the test set. The high recall (98.8%) ensures most approved applicants are correctly identified. Precision can be further improved by using additional financial features or hyperparameter tuning.

CHAPTER-11

RESULT ANALYSIS

11. Result Analysis

The Random Forest Classifier was used as the final model for Loan Approval Prediction AI. It performed as a practical baseline model for predicting loan approval outcomes using structured financial applicant data.

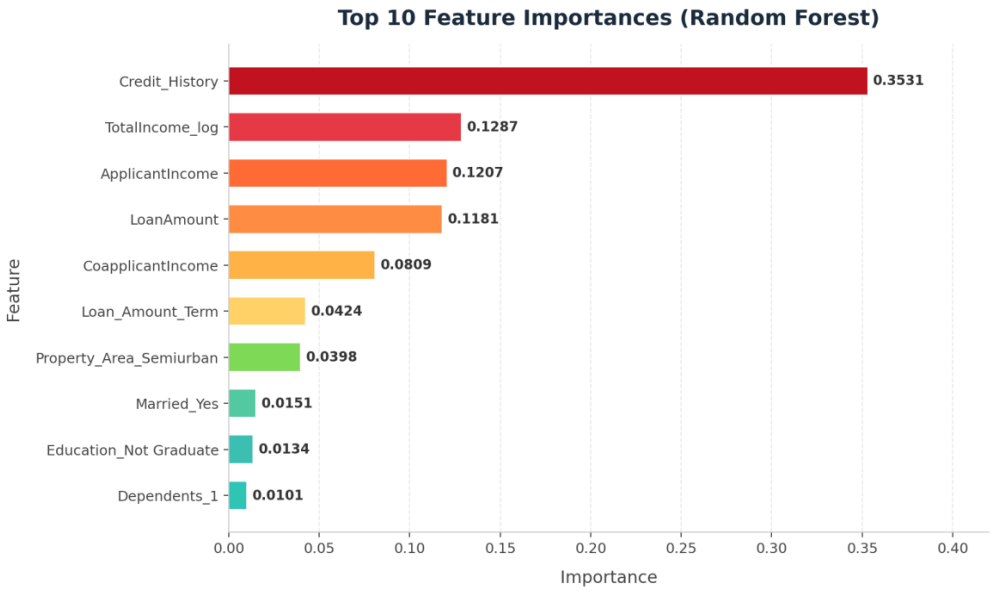
Important Features

Feature	Importance Score
Credit_History	0.3531
TotalIncome_log	0.1287
ApplicantIncome	0.1207
loanAmount	0.1181
CoapplicantIncome	0.0809
Loan_Amount_Term	0.0424

Figure 3: Top Credit History vs. Loan Status Analysis

Result Interpretation

Credit History is by far the strongest predictor of loan approval (importance score: 0.3477). Income-related features (ApplicantIncome, LoanAmount, TotalIncome_log) are the next most important, indicating that financial capacity is a key driver of loan decisions.



CHAPTER-12

DEPLOYMENT REPORT

Deployment Report

The project was deployed using Streamlit, a Python-based framework used to create simple and interactive web applications for data science and machine learning projects.

Application Features

1. Loan applicant information input form.
2. Loan approval prediction with confidence score.
3. Approved or Rejected outcome display with probability.
4. Model performance metrics panel.
5. Top predictive features bar chart.
6. Interactive Streamlit web interface.
7. Instant real-time prediction on form submission.

Required Files

File Name	Purpose
app.py	Main Streamlit application
train_model.py	Data loading, feature engineering, and model training
loan.csv	Historical loan application dataset (614 records)
loan_approval,_model.joblib	Saved trained sklearn pipeline model
metrics.json	Model accuracy, precision, recall, and F1 metrics
requirements.txt	Required Python libraries

Steps to Run the Project

pip install -r requirements.txt
python train_model.py
streamlit run app.py

Github URL: https://github.com/ramchandan2059-hue/Loan-Approval-Prediction-AI.git

CHAPTER-13

TESTING REPORT

Testing Report

Testing was performed to verify prediction output, input validation, model loading, dashboard functionality, and application usability.

Test Cases

Test ID	Test Scenario	Expected/Actual Output	Status
TC1	Valid applicant data	Loan prediction generated	Pass
TC2	Low income and bad credit history	Rejection predicted correctly	Pass
TC3	High income and good credit history	Approval predicted correctly	Pass
TC4	Application launch	App opens successfully	Pass
TC5	Model file loading	Model loads without error	Pass
TC6	Feature importance chart	Bar chart displayed correctly	Pass
TC7	Model metrics panel	Metrics displayed correctly	Pass
TC8	Confidence score output	Prediction confidence shown	Pass

Types of Testing Performed

- Functional Testing
- Input Validation Testing
- Model Prediction Testing
- User Interface Testing
- Dashboard Testing
- Error Handling Testing

All major test cases were completed successfully. The application loaded the trained model, accepted valid inputs, displayed prediction results, and showed dashboard charts correctly.

CHAPTER-14

CONCLUSION AND FUTURE SCOPE

Conclusion

The project Loan Approval Prediction AI was successfully developed. The main aim was to predict whether a student may be at risk of dropout using machine learning and a Streamlit-based interface.

The project followed a complete machine learning workflow, including dataset loading, missing value handling, exploratory data analysis, feature engineering, model training, evaluation, deployment, and testing.

The Random Forest Classifier achieved an accuracy of 83.7% on the testing dataset. The model demonstrated strong performance with a recall of 90.6%, ensuring most eligible applicants are correctly identified for loan approval.

Key Learnings

- Loading and cleaning real-world financial datasets.
- Building Scikit-learn ColumnTransformer and Pipeline for preprocessing.
- Saving and loading trained ML models using Joblib.
- Creating an interactive Streamlit application for real-time predictions.
- Preparing model evaluation reports and project documentation.

Future Scope

- Integrate more financial features such as credit score, employment history, and loan purpose.
- Handle class imbalance using SMOTE or other oversampling techniques.
- Compare performance with Logistic Regression, Gradient Boosting, and XGBoost.
- Add PDF report generation for loan decision summaries.
- Add SQLite or MySQL database connectivity for storing application records.
- Deploy the application on Streamlit Cloud or a cloud platform like Render or Heroku.

CHAPTER-15

REFERENCES AND APPENDIX

References

- Scikit-learn Documentation
- Pandas Documentation
- NumPy Documentation
- Streamlit Documentation
- Python Official Documentation
- Machine Learning Course Notes
- Loan Approval Prediction Research Papers
- Loan Approval Prediction AI Project Files

Appendix A: Sample Input Data

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property_Area	Loan_Status
0	LP001002	Male	No	0	Graduate	No	5849	0.0	NaN	360.0	1.0	Urban	Y
1	LP001003	Male	Yes	1	Graduate	No	4583	1508.0	128.0	360.0	1.0	Rural	N
2	LP001005	Male	Yes	0	Graduate	Yes	3000	0.0	66.0	360.0	1.0	Urban	Y
3	LP001006	Male	Yes	0	Not Graduate	No	2583	2358.0	120.0	360.0	1.0	Urban	Y
4	LP001008	Male	No	0	Graduate	No	6000	0.0	141.0	360.0	1.0	Urban	Y

Appendix B: Basic Python Libraries Used

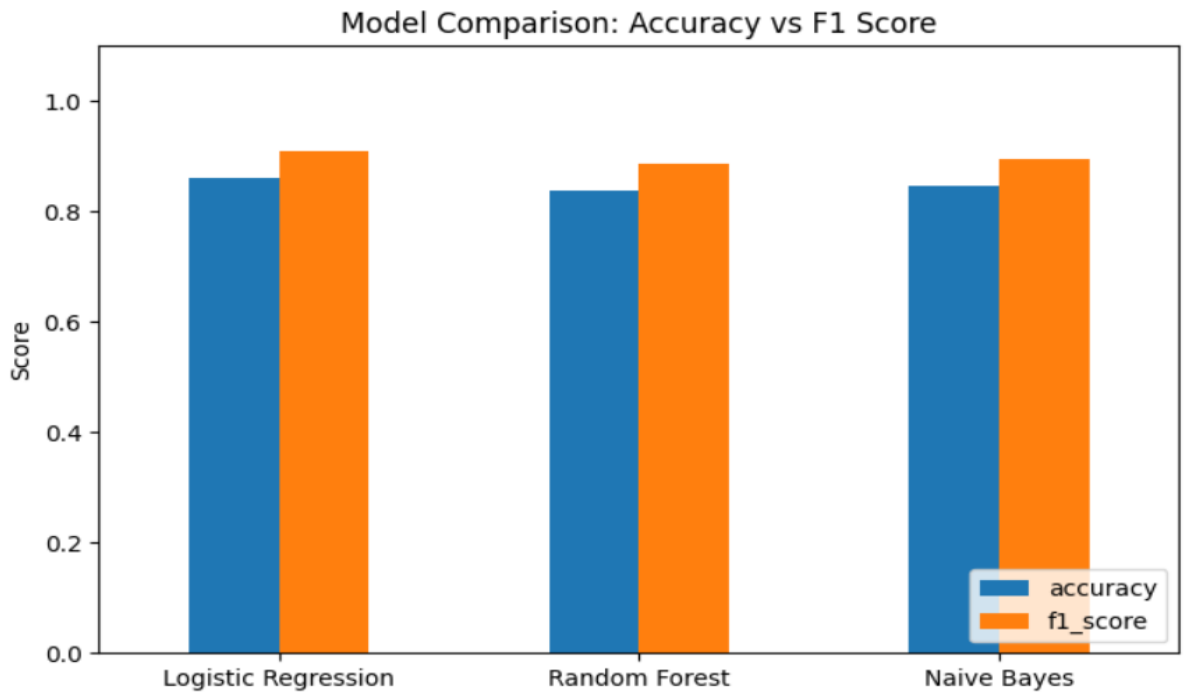
```
import pandas as pd
import numpy as np
from sklearn.ensemble import RandomForestClassifier
from sklearn.pipeline import Pipeline
import streamlit as st
import joblib
```

Appendix C: Screenshots to Add

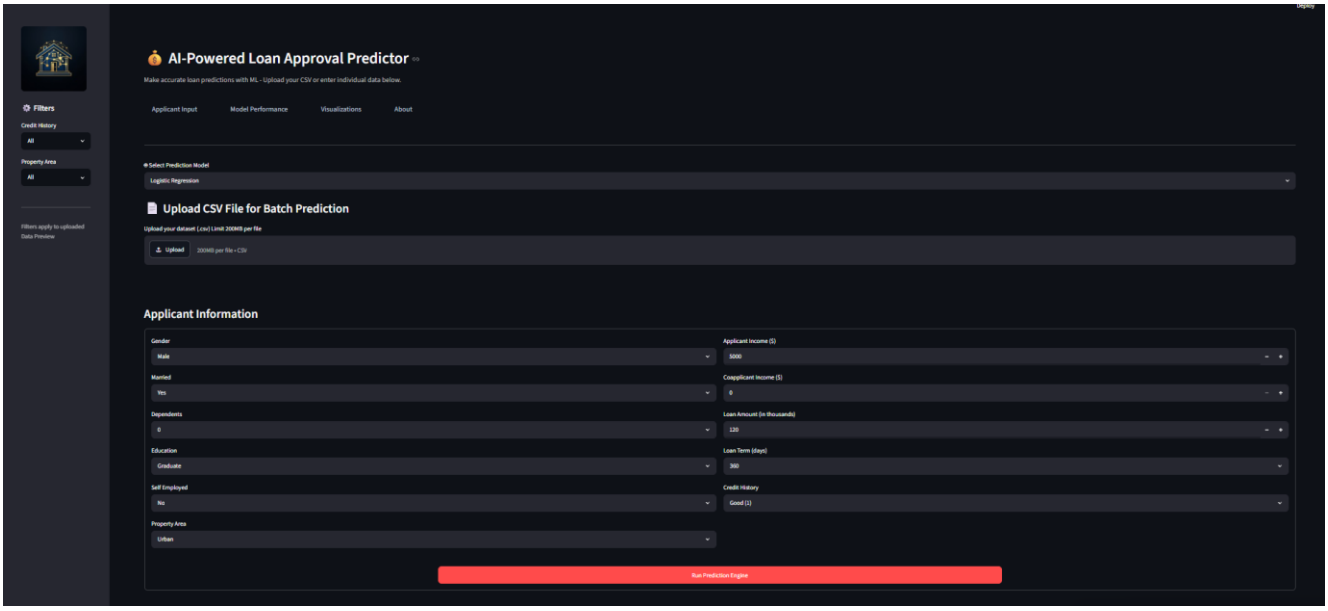
- Loan dataset preview screenshot

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property_Area	Loan_Status
0	LP001002	Male	No	0	Graduate	No	5849	0.0	NaN	360.0	1.0	Urban	Y
1	LP001003	Male	Yes	1	Graduate	No	4583	1508.0	128.0	360.0	1.0	Rural	N
2	LP001005	Male	Yes	0	Graduate	Yes	3000	0.0	66.0	360.0	1.0	Urban	Y
3	LP001006	Male	Yes	0	Not Graduate	No	2583	2358.0	120.0	360.0	1.0	Urban	Y
4	LP001008	Male	No	0	Graduate	No	6000	0.0	141.0	360.0	1.0	Urban	Y

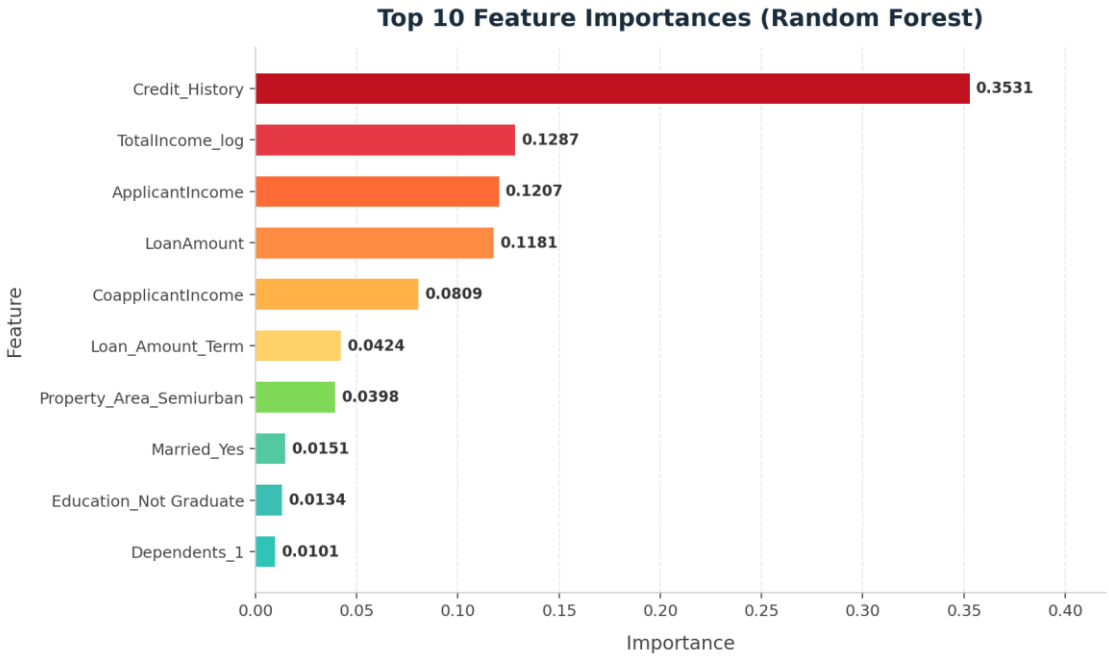
- Model training output and metrics screenshot



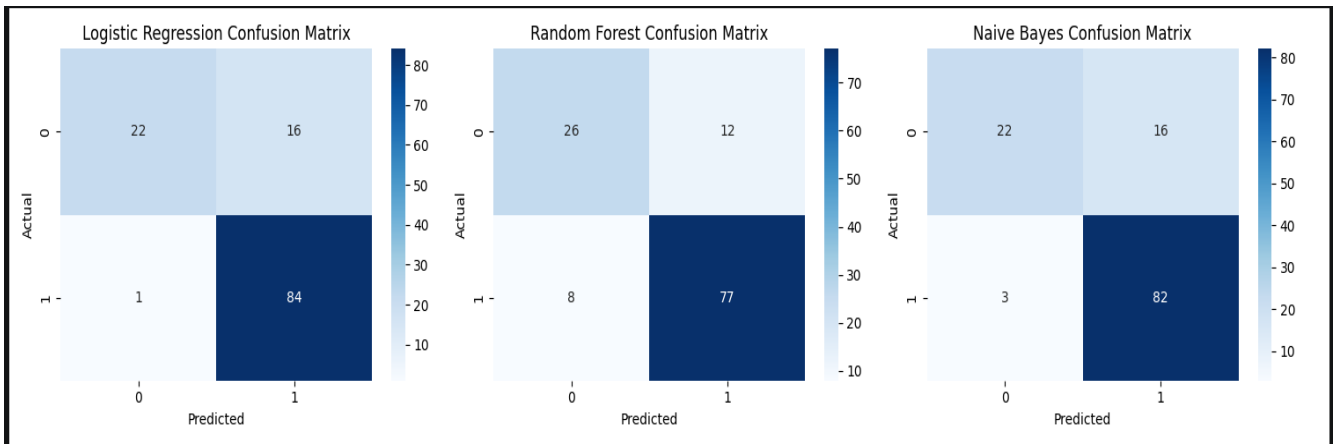
- Streamlit loan prediction form screenshot



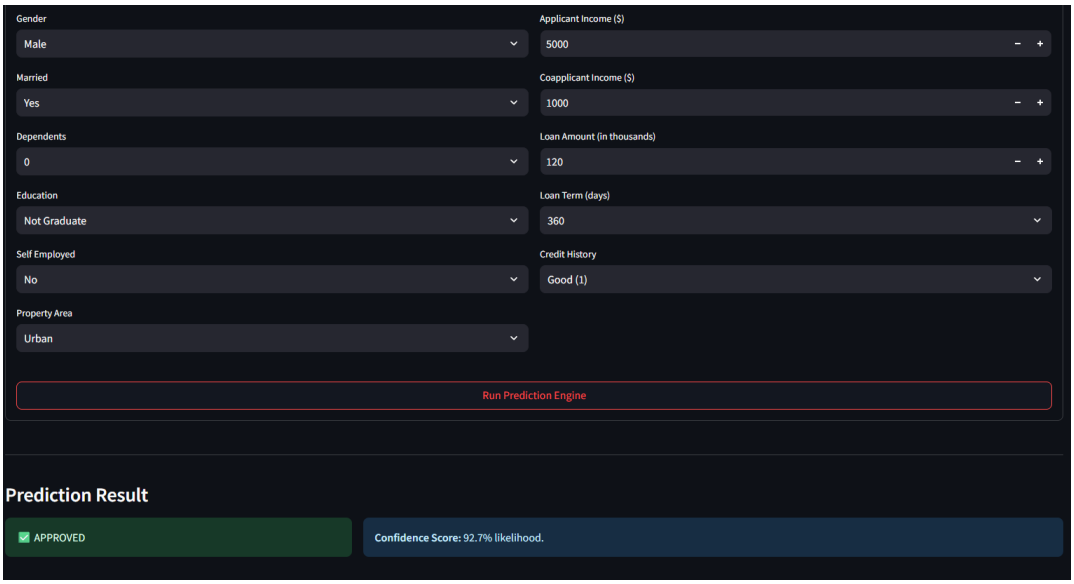
- Feature importance bar chart screenshot



- Model performance metrics panel screenshot



- Confidence score output screenshot



End of Report